

Solve this equation:

$$e^{2x} - 5e^x + 6 = 0$$

$$\text{Let } t = e^x$$

$$t^2 - 5t + 6 = 0$$

$$t_1 = 3 \quad \text{and} \quad t_2 = 2$$

$$e^x = 3$$

$$e^x = 2$$

$$x = \ln 3$$

$$x = \ln 2$$

$$S: (\ln 3, \ln 2)$$



Solve the equation:

$$\left(\frac{2}{3}\right)^x = \frac{9}{4}$$

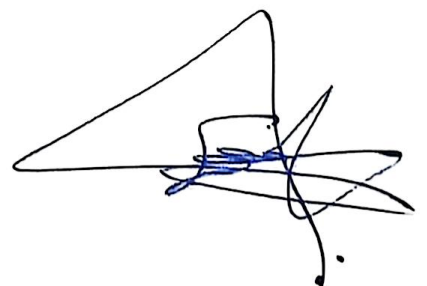
Solution

$$\left(\frac{2}{3}\right)^x = \left(\frac{3}{2}\right)^2$$

$$\left(\frac{2}{3}\right)^x = \left(\frac{2}{3}\right)^{-2}$$

$$\text{So: } x = -2$$

$$S = \{-2\}$$



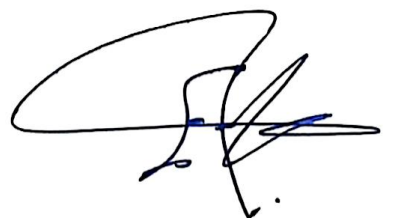
Solve the following equation:

$$7^{x-5} = 1$$

We know $7^0 = 1$

So: $x-5 = 0$
 $x = 5$

S: $\{5\}$



Solve the following equation:

$$9^x - 4 \cdot 3^x + 3 = 0$$

$$\text{Let } t = 3^x$$

$$9^x = (3 \cdot 3)^x = 3^x \cdot 3^x = 3^{2x}$$

$$t^2 - 4t + 3 = 0$$

$$t_1 = 3$$

and

$$t_2 = 1$$

$$3^x = 3$$

$$3^x = 1$$

$$\text{so: } x = 0$$

$$\text{so: } x = 1$$

+

$$\therefore \{1, 0\} \quad \text{H}$$



Solve the following equation.

$$2^{3x-2} = 4^{x+1}$$

$$4^{x+1} = (2 \cdot 2)^{x+1} = 2^{x+1} \cdot 2^{x+1} = 2^{2x+2}$$

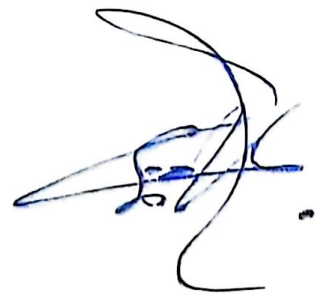
$$\Rightarrow 2^{3x-2} = 2^{2x+2}$$

$$\text{then: } 3x-2 = 2x+2$$

$$3x-2x = +2+2$$

$$x = 4$$

$$S: \{4\}$$



Solve the following equation:

$$a) 2^{x+1} = 16$$

Solution:

$$\text{We know } 2^4 = 16$$

$$2^{x+1} = 2^4 \Rightarrow x+1 = 4$$
$$\boxed{x = 3}$$

$$\therefore \{3\}$$

$$b) 3^{2x+1} = 27$$

Solution:

$$\text{We know } 27 = 3^3$$

$$\text{So: } 3^{2x+1} = 3^3 \Rightarrow 2x+1 = 3$$
$$2x = 2$$
$$\boxed{x = 1}$$

$$\therefore x = 1$$

$$c) 5^{x^2-1} = 125$$

Solution

$$\text{We Know: } 5^3 = 125$$

$$5^{x^2-1} = 5^3 \Rightarrow x^2-1 = 3$$

$$x^2 = 4$$

$$x = \sqrt{4} = 2$$

$$x = \pm 2$$

$$d) 4^x = \frac{1}{64}$$

$$4^x = 64^{-1}$$

$$\text{We Know: } 64 = 4^3 \Rightarrow 4^x = \frac{1}{4^3}$$

$$\Rightarrow 4^x = 4^{-3}$$

$$\text{then, } x = -3$$

$$x = -3$$



Solve this legendary equation:

$$10^{x+3} = 0.001$$

Solution:

$$\text{We know: } 0.001 = 10^{-3}$$

$$\text{So: } 10^{x+3} = 10^{-3}$$

$$\Rightarrow x+3 = -3$$

$$x = -6$$

$$S: \{-6\}$$

E-Vex is building...

